Ejercicio 2-6

miércoles, 1 de abril de 2026 21:51

b) $BC + A\bar{C} + AB + BCD$

$BC(1+D) + A(\bar{C}+B)$

$B(A+CD) + BC + A\bar{C}$

$[AB + BC + A\bar{C}] + BCD$

$AB[1 + C + \bar{C}] + BCD$

$AB + BCD$ no aplica a las leyes

$B[A + CD]$

REDUCIR A CUATRO LITERALES

$[BC + BCD] + A\bar{C} + AB$

$BC + A\bar{C} + AB[C + \bar{C}]$

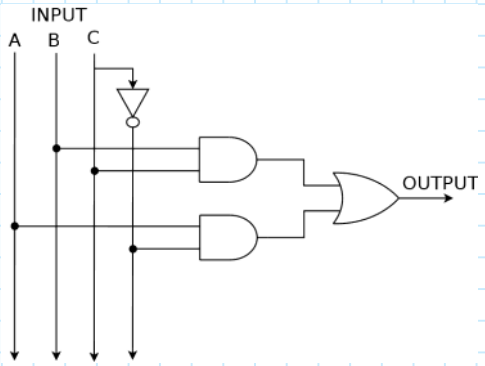
$BC + A\bar{C} + ABC + AB\bar{C}$

$BC(1+A) + A\bar{C}(1+B)$

$BC + A\bar{C}$

$\boxed{R11 \quad BC + A\bar{C}}$

DIAGRAMA



c) $[(CD)' + A]' + A + CD + AB$

$[C' + D' + A]'$ MORGAN

$C + D + A' + A + CD + AB$

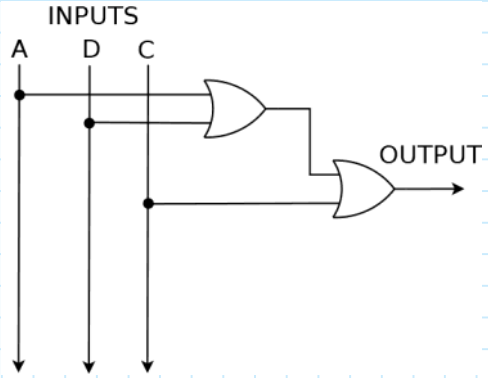
$A(1+B) + C + D + CD$

$A(1+B) + D + (C + CD)$ LEY DE ABSORCIÓN

$A(1+B) + D + C$ LEY DE ABSORCIÓN

$\boxed{A + D + C}$

A TRES LITERALES



$\boxed{d}$ $(A+C+D)(A+C+D')(A+C'+D)(A+B')$ ⟶ A CUATRO LITERALES

$[(A+C)+D)((A+C)+D')]$ ✗

$[A+C+D'](A+C'+D)(A+B')$

$(A+C'+D')(A+B')$

$A[(C'+D')(B')]$

$[A+C](D+D')$ $A[(C'+D)(1+B')]$ ✗

$(A+C)(1+1)$ $A[(C'+D)(1+B')]$

$A[(1+C+D)(1+C+D')(1+\bar{C}+D)(1+\bar{B})]$

$(A+C+D)(A+C+D')(A+C'+D)(A+B')$

$AA + AC + AD' + AC + CC + CD' + AD + CD + \cancel{DD'}$

$(A + AC) + AD' + (AC + C) + CD' + AD + CD$ ✓

A C

$A + C + A\bar{D} + AD + C\bar{D} + CD$

$A + C + A(\cancel{\bar{D}+D})^{1} + C(\cancel{\bar{D}+D})^{1}$

$A + C + A + C = A + C$